Mini Project Presentation

HostelPulse

Hostel Issue Management System

GROUP MEMBERS

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2025–2026 | IET Lucknow, CSE-R

Outline

Contents

1. Introduction
2. Problem Statement & Objectives
3. System Flow & Architecture
4. Technology Stack
5. Website Screenshots
6. Database Design
7. Conclusion
8. References

HostelPulse digitalizes hostel complaint management, replacing paper-based workflows with a transparent, role-based online system.

- Students submit & track issues
- Admin manages & resolves them
- Deployed on AWS
- MongoDB Atlas cloud database

Introduction

What is HostelPulse?

A **web-based platform** built to streamline reporting and resolution of hostel-related issues for students and administrators.

Students raise complaints online and track their status in real time, while wardens manage and resolve them through a centralized dashboard.

Target Users: Students Admin/Warden

Key Highlights

- Online complaint submission portal
- Role-based access (Student / Admin)
- Real-time issue status tracking
- Admin dashboard for management
- JWT-secured authentication
- MongoDB cloud database
- Deployed fully on AWS

Problem Statement & Objectives

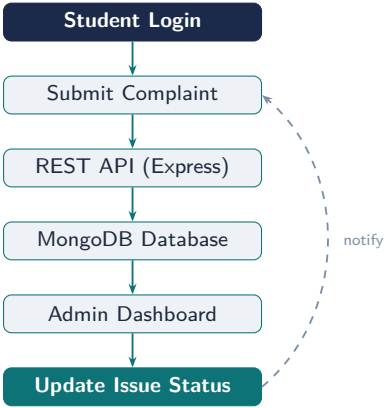
Problems

- Complaints reported manually on paper
- No centralized tracking system
- No accountability for resolution
- Students unaware of complaint status
- Difficult to identify recurring issues
- Warden overload from verbal complaints

Objectives

- Build a digital complaint submission portal
- Provide real-time status updates to students
- Enable admin to manage issues centrally
- Implement role-based secure authentication
- Reduce manual workload through automation

System Flow & Architecture



Student Actions

- Register and login securely
- Submit hostel complaints
- Track complaint status
- Edit or delete own issues

Admin / Warden Actions

- View all complaints
- Update issue status

Technology Stack

Frontend

React.js
Component-based UI

Tailwind CSS
Utility-first styling

Axios
HTTP client

Backend

Node.js
JS runtime

Express.js
REST API framework

JWT + bcrypt
Auth & hashing

Database

MongoDB Atlas
Cloud NoSQL DB

Mongoose
ODM for Node

Schema Validation
Strict type checks

Deployment

AWS
Frontend & Backend

Git / GitHub
Source control

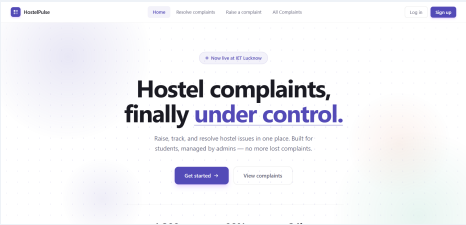
MongoDB Atlas
Cloud database

Repo: `https://github.com/ietlucknowofficial/HostelPulse` **Hosted on AWS**

Website Screenshots

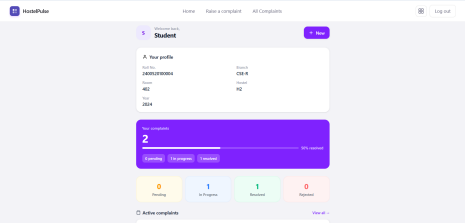
Home & Student View

Home Page



Landing page with login & registration

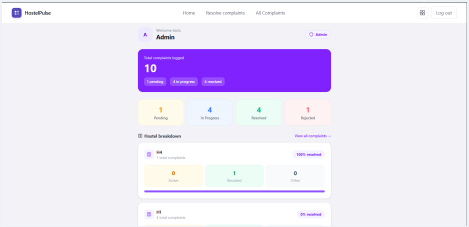
Student Dashboard



Student complaint tracking dashboard

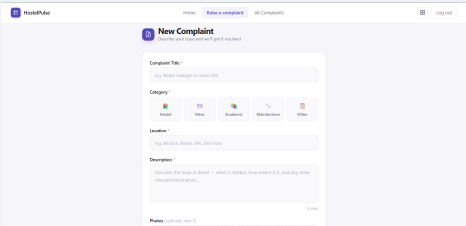
Website Screenshots

Admin Dashboard



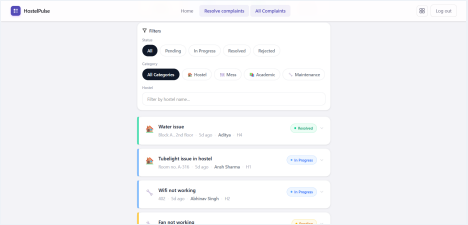
Centralized admin panel with statistics

Submit Complaint Form



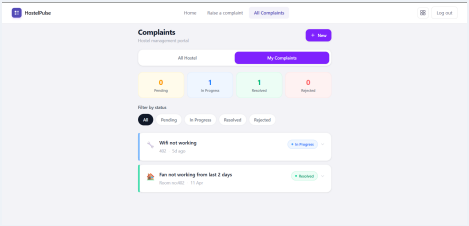
Structured form with category & room details

Complaints List (Admin)



Admin view: manage all student issues

Complaints List (Student)

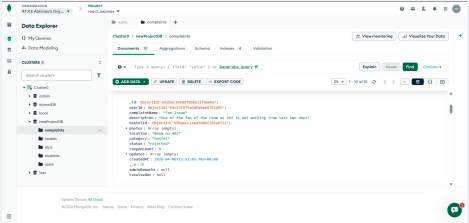


All complaints submitted by a student

Website Screenshots

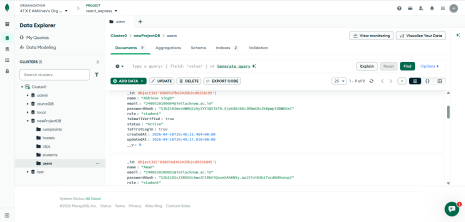
Complaint Database & User Collection

Complaint Database



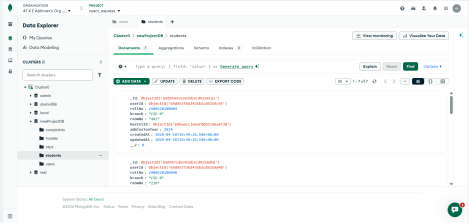
Complaints collection in MongoDB Atlas

User Collection



Users collection in MongoDB Atlas

Students Collection



Students collection stored in MongoDB Atlas

Collection Fields

users collection:

- `_id, name, email`
- `password` (bcrypt hashed)
- `role: student | admin`
- `rollNo, roomNo, isVerified`

complaints collection:

- `_id, userId` (ref to user)
- `title, description, category`
- `roomNo`
- `status: pending | resolved`
- `createdAt, updatedAt`

Conclusion

Summary

HostelPulse successfully digitalizes the hostel complaint process with a secure, scalable MERN application. It provides:

- Faster complaint resolution
- Transparent communication
- Reduced manual workload
- Centralized hostel management
- Better monitoring of recurring issues

Future Enhancements

- Push notifications (email / SMS)
- Analytics dashboard with charts
- Mobile app (React Native)
- Multi-hostel / multi-campus support
- Complaint priority levels
- Scheduled maintenance alerts

https://github.com/ietlucknowofficial

https://d1fwtw1io6mluw.cloudfront.net

References

Documentation & Resources

1. React.js Official Docs — `react.dev`
2. Node.js Documentation — `nodejs.org/docs`
3. Express.js Guide — `expressjs.com`
4. MongoDB Atlas Docs — `mongodb.com/docs`
5. JWT Standard (RFC 7519) — `jwt.io`
6. AWS Documentation — `docs.aws.amazon.com`
7. Tailwind CSS Docs — `tailwindcss.com`

Project Links

GitHub Repository:

`https://github.com/ietlucknowofficial/`

Hosted on AWS:

`https://d1fwtw1io6mluw.cloudfront.net/`

Built with the MERN Stack

IET Lucknow, CSE-R

Academic Year 2025–26

Thank You!

Questions & Discussion

Abhinav Singh ▪ Ansh Sharma ▪ Bhavya Nigam
Devraj Gupta ▪ Himanshu

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